

Charles Zhao

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WORK EXPERIENCE

Lawrence Livermore National Laboratory

Summer 2023, 2024

Cybersecurity Intern, 40 hrs/wk

Livermore, CA

- Improved development efficiency for lead developers by building unit and integration tests using Python (sqlite3, psycpg2, pytest), reducing debugging time and ensuring CI/CD pipeline compatibility within an Agile team environment.
- Built a dynamic web application for the Department of Homeland Security using the MEAN stack (MongoDB, Express, Angular, Node), delivering a scalable and efficient solution for enhanced data management.
- Developed a Python script to automate report parsing and identify suspicious IP addresses and domains interacting with the network, improving threat detection and security monitoring.

PROJECTS

KubeRange

April 2026

- Developed a full-stack web application (React, FastAPI, PostgreSQL) hosted on Kubernetes that dynamically provisions and tears down isolated lab environments (DVWA, Juice Shop) on demand via the Kubernetes API, with integrated ELK stack SIEM for security monitoring.

Cascading Detection Engine

April 2026

- Built a cascading phishing email detection engine with rule-based, classical ML (SVM), unsupervised (Isolation Forest), and transformer (DistilBERT) layers that escalate ambiguous samples to the next stage, with a Logistic Regression fusion layer for final scoring.

Cybersecurity Daily Digest

April 2026

- Built an automated cybersecurity news pipeline using RSS feeds, GitHub Actions, and an LLM to collect, summarize, and deliver a daily digest.

Domain Sinkhole & Network Traffic Analysis

December 2025

- Deployed a local DNS sinkhole using Pi-hole and Raspberry Pi to filter and block unwanted domains across all devices on a home network. Investigated captured DNS queries using Wireshark and documented my findings on GitHub.

Custom C++ Stager

May 2025

- Developed a custom C++ stager within a sandboxed Windows environment to study and evaluate modern antivirus evasion techniques, including process injection and dynamic API resolution.

Malicious File/URL Scanner

May 2024

- Designed and delivered a Python-based scanner for extracted attachments and URLs, leveraging VirusTotal's API to identify and flag malicious content in emails.

EDUCATION

Virginia Tech

- *Master of Engineering, Computer Engineering*

May 2025

- GPA: 4.0

- *Bachelor of Science, Computer Engineering*

May 2024

- GPA: 3.61; *Magna Cum Laude*; IC CAE; SFS CyberCorps

CERTIFICATIONS, SKILLS & INTERESTS

- **Certifications:** CompTIA Security+; Cranium AI Red Team Certification
- **Skills:** Python; C++; Testing & Automation; Wireshark; Burp Suite; Linux
- **Interests:** AI infrastructure & supply chains; Drone photography; Dog training; Strength training